

MATH9102

DIFFERENCE – CONTINUOUS VARIABLE – 2 GROUPS

Sources used in creation of this lecture:

Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using R
Field, Miles and Field; Understanding Basic Statistics, Brase and Brase; SPSS Survival
Manual, Julie Pallant

Before Inferential Statistics...

We need to establish evidence to support going ahead with building a predictive model

If we are asserting a relationship

- We need to investigate if there is any evidence of a relationship using the appropriate test and make a decision based on the results (strength, direction etc.)

If we are asserting a differential effect for different groups

- We need to investigate if there is any difference using the appropriate test and make a decision based on the result

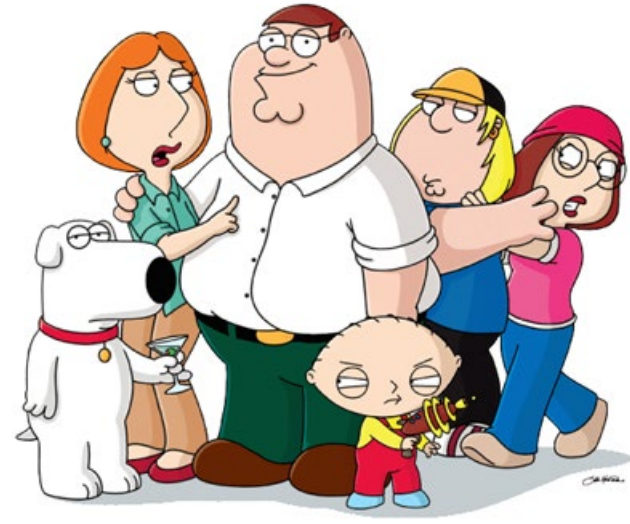


Sometimes we want to investigate if different groups within our population experience a certain concept differently, or if a specific effect varies across these group

There are a range of statistical tests we can use. The choice depends on:

- The measurement of the variable

- The shape of the data

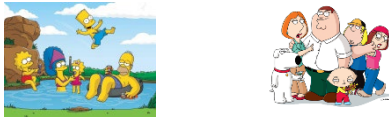
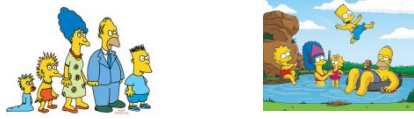


Are these groups different?

"Is there a significant difference in the average amount of time spent on family activities per episode between the Griffin family and the Simpsons family?"

Comparison between Samples

Tests for Group Comparison

	Independent Samples 	Related Samples also called dependent means test 
Interval measures/ parametric	Independent samples t-test*	Paired samples t-test**
Ordinal/ non-parametric	Mann-Whitney U-Test*	Wilcoxon test **

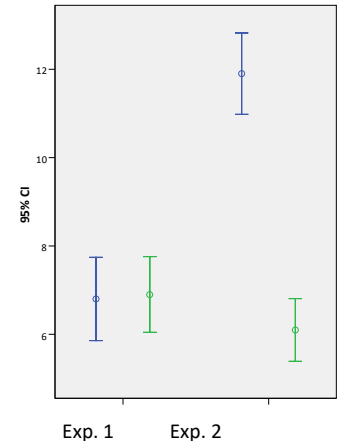
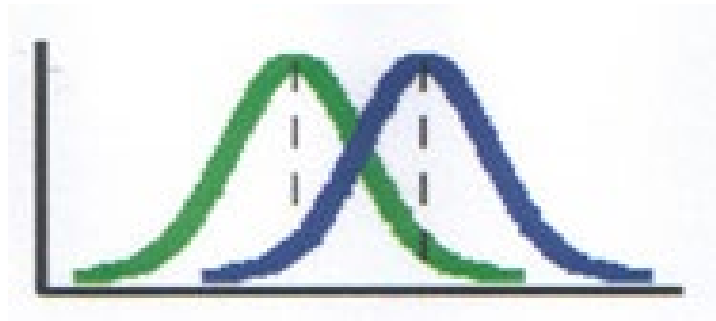
* 2 different groups of participants

** 2 same participants measured at two different points

Parametric

MATH9102-W4-TTEST.QMD

Parametric: t-tests



Compare the **mean** between 2 samples/groups/conditions

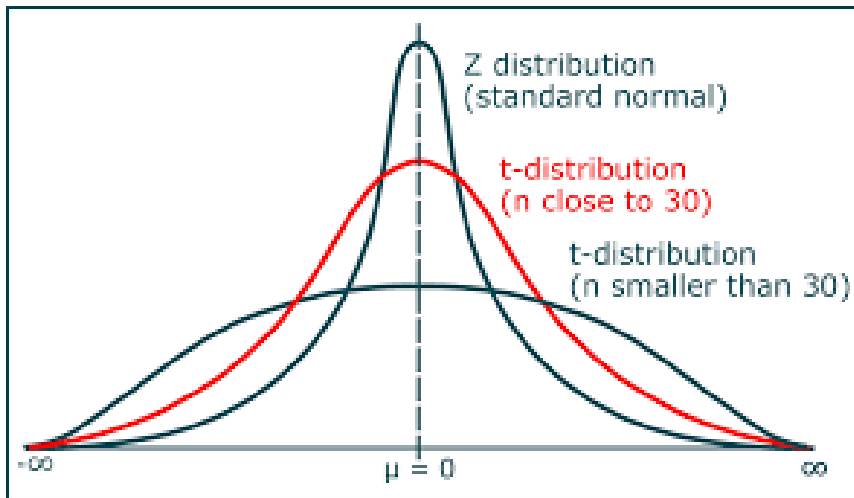
If 2 samples are taken from the same population

- Then they should have similar means

If 2 means are statistically different

- Then the samples are likely to be drawn from 2 different populations,
 - i.e. they really are different

t Distribution



Very similar to the Z distribution by assuming normality.

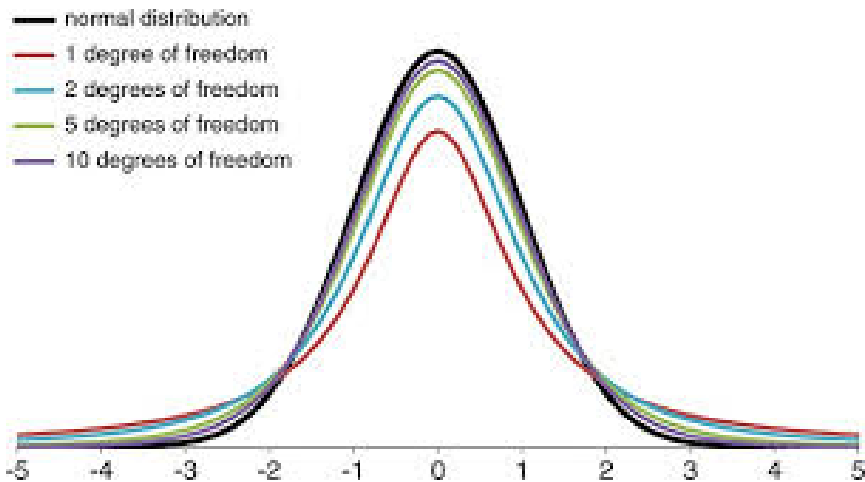
Normality is obtained after about 120 data observations.

- In which case $t = z$

Basic rule of parameter estimation:

- The higher the observations (N) of sample the more reflective of overall population.

The t Distribution



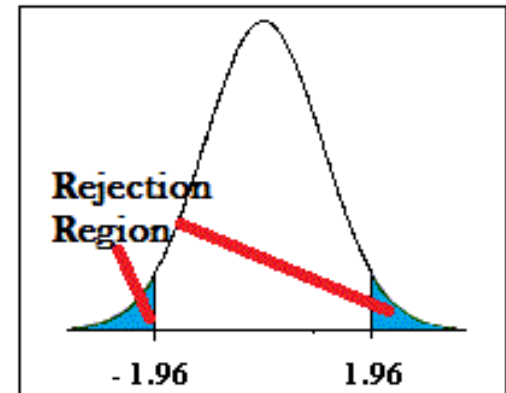
The shape of t depends on its degrees of freedom (number of observations used to calculate the statistic).

As df becomes infinitely large, t becomes normal and becomes z

t critical values table

https://www.tutorialspoint.com/statistics/t_distribution_table.htm

Your statistical tool will do this look up for you and calculate whether your test is statistically significant or not.



Formula for test statistic

Difference between the means divided by the pooled **standard error of the mean**

$t = \frac{M_1 - M_2}{S_{\text{pooled}}}$	Mean of group 1 (M_1) minus mean of group 2 (M_2), divided by the pooled standard error (S_{pooled}).
$S_{\text{pooled}} = \sqrt{\left(\frac{\hat{s}_1}{\sqrt{N_1}}\right)^2 + \left(\frac{\hat{s}_2}{\sqrt{N_2}}\right)^2}$	Standard deviation of group 1 ($\hat{s}_1$) divided by the square root of the number of people in group 1 ($\sqrt{N_1}$), squared (2); plus standard deviation of group 2 ($\hat{s}_2$) divided by the square root of the number of people in group 2 ($\sqrt{N_2}$), squared (2); then square root of everything ($\sqrt{}$).

SE variability between sample means

If large, then may be a problem with representativeness of the sample

Formula for test statistic

<https://www.peggykern.org/uploads/5/6/6/7/56678211/ttestformulas.pdf>

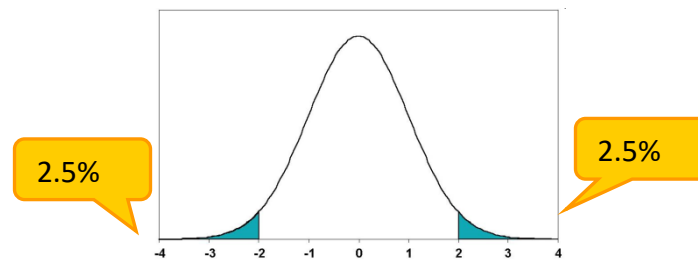
- Dr. Peggy (Margaret) L. Kern, University of Melbourne

Very useful – all the formulas for different types of t-test with explanation

Types of t-tests

2-tailed tests

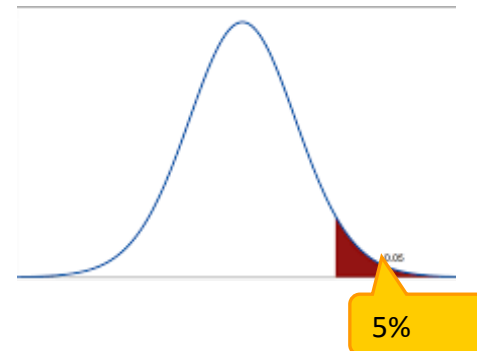
- **H0 (Null):** There is no difference in perceived stress for respondents who have children and those respondents who do not
- **Ha (Alternate):** There is a difference in perceived stress for respondents who have children and those respondents who do not
- **What does this mean:**
 - $H_0: \mu \text{ parents} = \mu \text{ non-parents}$
 - $H_a: \mu \text{ parents} \neq \mu \text{ non-parents}$



Types of t-tests

1-tailed tests

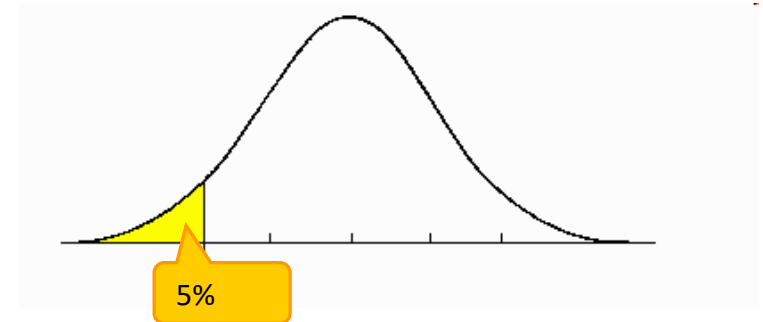
- **E.g.**
 - **H₀ (Null)**: Parents do not have higher perceived stress levels than non-parents.
 - **H_a (Alternate)**: Parents have higher perceived stress levels than non-parents.
 - **What does this mean:**
 - H₀: $\mu_{\text{parents}} \leq \mu_{\text{non-parents}}$
 - H_a: $\mu_{\text{parents}} > \mu_{\text{non-parents}}$



Types of t-tests

1-tailed tests

- E.g.
 - **H0 (Null)**: Parents do not have lower perceived stress levels than non-parents.
 - **Ha (Alternate)**: Parents have lower perceived stress levels than non-parents.
 - **What does this mean:**
 - $H_0: \mu_{\text{parents}} \geq \mu_{\text{non-parents}}$
 - $H_{1a}: \mu_{\text{parents}} < \mu_{\text{non-parents}}$



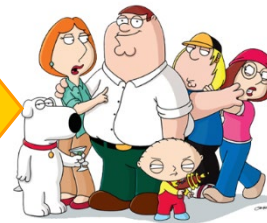
Types of t-tests

2 sample (groups) t-tests

- Takes your sample data from two groups



Mean



Mean

1 sample (group) t-tests



Mean



A known
value

Assumptions of the t -test

Both the independent t -test and the dependent t -test are *parametric tests* based on the normal distribution.

Therefore, they assume:

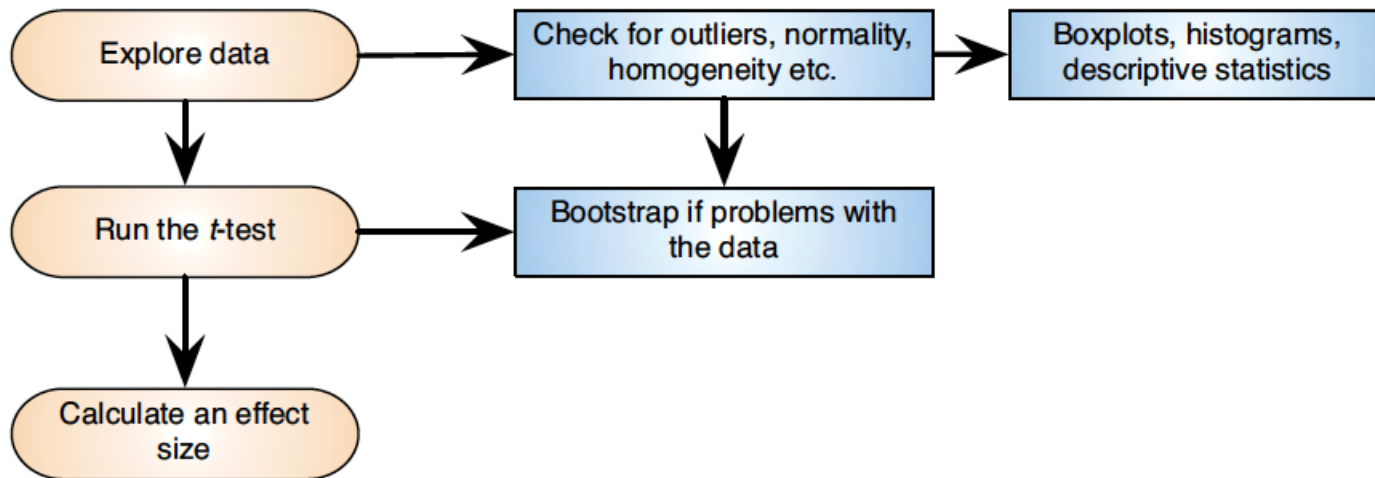
- Data are measured at least at the interval level (interval or ratio data is acceptable).
- The sampling distribution is normally distributed.

Assumptions of the t -test

And because it is used to test different groups, also assumes:

- Variances in these populations are roughly equal (*homogeneity of variance*).
- Scores in different treatment conditions are independent (because they come from different people).

Independent t -test process



Independent *t*-test Example

Dataset:

- Survey.dat (Julie Pallant)

Question:

- Is there a significant difference in the mean perceived stress scores for respondents who have children and respondents who do not have children?

Hypothesis (Two-tailed):

- H₀: There is no difference in perceived stress scores for respondents who have children and respondents who do not have children
- H_A: There is a difference in perceived stress scores for respondents who have children and respondents who do not have children

Cut-off or α level:

- 0.05

T-test:

- Will tell you whether there is a statistically significant difference between the mean scores of the groups

Independent t-test in R

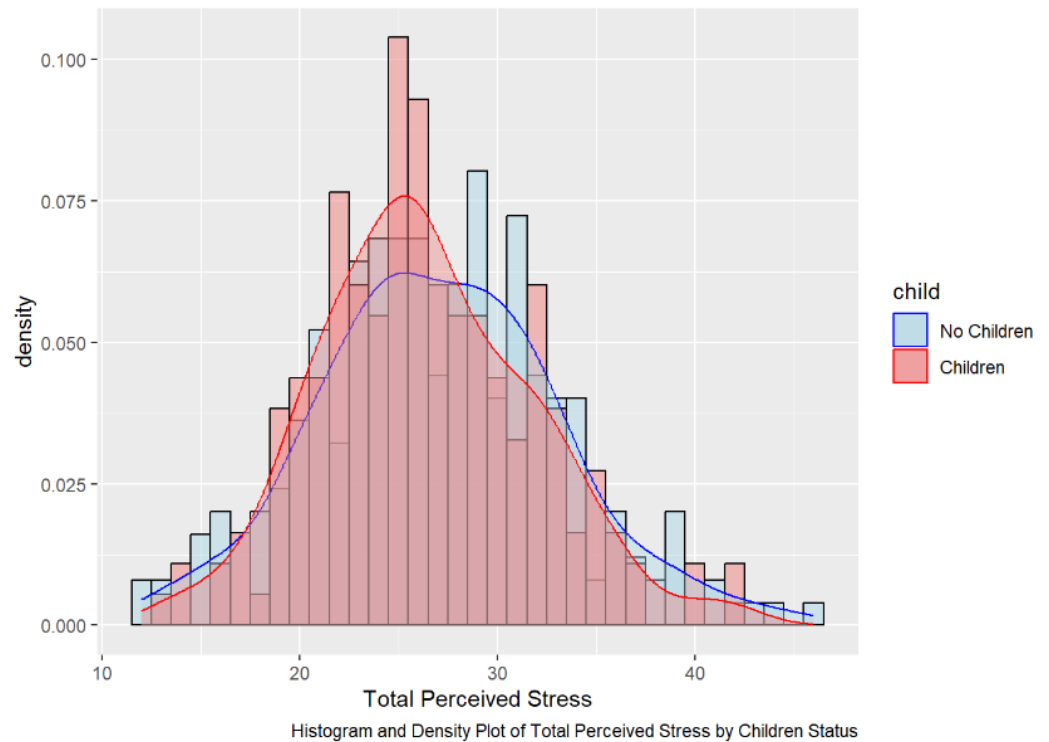
Step 1: Assess and Describe the scale variable first

- **As per previous examples**
 - Histogram with density curve (or boxplot)
 - QQPlot
 - Standardised skew and kurtosis
 - Percentage of absolute scores falling within bounds
 - Appropriate central tendency and dispersion statistics

Independent t-test in R

Step 1: Assess and Describe the scale variable first

- Look at each group
 - Histogram with density curve (or boxplot)
 - We only need to do the skewed and kurtosis check on the variable as a whole



Independent t-test in R

**Step 2: Assess
the groups and
describe the
scale variable
per group**

We need to know how many are in each group.

- Percentages or contingency table
- Bar chart or pie chart

Independent t-test in R

Step 2: Assess the groups and describe the scale variable per group

We know from last week that we can treat the data for these two variables as a normal distribution

We need to look at the descriptive statistics for the variable by group

M=27.04, SD= 6.133 for respondents who have children

M= 26.34, SD= 5.43 those for who don't

Independent t-test in R

Step 3: Test for Homogeneity of Variance

Step 3: Test for homogeneity of variance in your groups

In R you need to choose whether to run a t-test with modification.

This requires you to test homogeneity of variance in advance of running other tests

- Levene's test is best – this is the most robust test for normally distributed data. Null hypothesis is that the variance is equal.
- To run in R
 - Levene's is in the library car

```
library(car)
```
 - ```
leveneTest(continuousvariable ~ categoricalvariable, data=dataframename)
```
  - Gives as output an F statistic and an estimate of significance
    - Don't worry about F for now we will talk next week

# Independent t-test in R Step 3

**Step 3:** We need to know the homogeneity of variance in advance so we can set the parameter on the t-test so we conduct the Levene's test first

```
#Conduct Levene's test for homogeneity of variance in library car - the null hypothesis is that variances in group
s are equal so to assume homogeneity we would expect probability to not be statistically significant.
car::leveneTest(tpstress ~ child, data=survey)
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

|       | Df<br><int> | F value<br><dbl> | Pr(>F)<br><dbl> |
|-------|-------------|------------------|-----------------|
| group | 1           | 3.184236         | 0.07505667      |
|       | 430         | NA               |                 |

The null hypothesis for Levene's test is that variances are equal.

If the resulting p-value is under 0.05 it means that variances are not equal, we need to employ a correction.

Note: this test is meant to be used with normally distributed data but can tolerate relatively low deviation from normality.

# Independent t-test in R

## Step 4 – Conduct the test

### Step 4: Conduct the t-test

**t.test( )** function produces a variety of t-tests.

- Unlike most statistical packages, the default assumes unequal variance and applies the Welch df modification.
- You can use the **var.equal = TRUE** option to specify equal variances and a pooled variance estimate.

```
t.test(y~x) # where y is numeric
and x is a binary factor
```

```
t.test (tslfest~child)
```

# Independent t-test in R

## Step 5: Interpret the findings

### Step 5: Interpret the findings

**t.test( )** function produces a variety of t-tests.

`t.test(y~x)` # where y is numeric and x is a binary factor

```
#Conduct the t-test from package stats
#In this case we can use the var.equal = TRUE option to specify equal variances and a pooled variance estimate
stats::t.test(tpstress~child,var.equal=TRUE,data=survey)
```

```
##
Two Sample t-test
##
data: tpstress by child
t = 1.2254, df = 430, p-value = 0.2211
alternative hypothesis: true difference in means between group NO and group YES is not equal to 0
95 percent confidence interval:
-0.4211406 1.8158341
sample estimates:
mean in group NO mean in group YES
27.03614 26.33880
```

In this case our p value is not less than our cut-off so it is not statistically significant.

## Step 5: Interpret the findings

- Look also at the mean difference

```
#Conduct the t-test from package stats
#In this case we can use the var.equal = TRUE option to specify equal variances and a pooled variance estimate
stats::t.test(tpstress~child,var.equal=TRUE,data=survey)
```

```

Two Sample t-test

data: tpstress by child
t = 1.2254, df = 430, p-value = 0.2211
alternative hypothesis: true difference in means between group NO and group YES is not equal to 0
95 percent confidence interval:
-0.4211406 1.8158341
sample estimates:
mean in group NO mean in group YES
27.03614 26.33880
```

Here we can see there is very little difference between the two groups – mean difference of 0.697

# Independent t-test in R

## Step 5

# Independent t-test in R

## Step 5

### Step 5: Interpret the findings

- Calculate an effect size

Looking to quantify the difference between the two groups

Cohen's d

- Calculated from t-statistic and the degrees of freedom.
- Guidelines on effect size:
  - 0.2=small effect
  - 0.5=moderate
  - 0.8 = large

OR

Eta Squared

- Magnitude of the difference between the means of your groups
- Guidelines on effect size:
  - 0.01 = small
  - 0.06 = moderate
  - 0.14 =large

NOTE: Choice of using eta squared or Cohen's d depends on your field of study

# Independent t-test in R

## Step 5

### Step 5: Interpret the findings

- Calculate an effect size

#### Cohen's $d$

- Value for a between subjects  $t$  test and the degrees of freedom.

$$\text{Cohen's } d = 2t / \sqrt{df}$$

In our case  $t=1.2254$ .,  $df = \# \text{ children}(183) + \# \text{ no children } (249) - 2 = 430$

$$2 * 1.2254 / \sqrt{430} = 0.059 - \text{rounds to } d = 0.06$$

#### Reporting Cohen's $d$ – convention:

- 0.2=small effect
- 0.5=moderate
- 0.8 = large

# Independent t-test in R

## Step 5

### Step 5: Interpret the findings

- Calculate an effect size

Eta squared

Magnitude of the difference between the means  
of your groups

Eta squared ranges from 0 to 1

- 0.01 indicates a small effect
- 0.06 indicates a medium effect
- 0.14 indicates a large effect

$$\text{eta squared} = \frac{t^2}{t^2 + (N1 + N2 - 2)}$$

For our example this is 0.003

# Independent t-test in R

## Step 6

### Step 6: Report your findings

#### Reporting convention:

- the statistic –  $t$
- Degrees of freedom (**df**) – number of pieces information used to calculate the statistic
- Probability of getting this statistic (**p**)
  - Non statistical significance result
    - Report the actual p value
  - Statistically significant result
    - Report  $p < 0.05$
    - Unless it is extremely significant
      - $p < 0.001$
- The effect size (even if not statistically significant)
  - Cohens  $d$  or eta squared
  - Comment on the effect size.

# Independent t-test in R

## Step 6

### Step 6: Report your findings

Non-statistically significant result:

An independent-samples t-test was conducted to compare perceived stress scores for respondents who have children and those who do not have children. Levene's test for equality of variances was not statistically significant, indicating that the assumption of homogeneity of variance was met ( $p > .05$ ). No significant difference in the scores for perceived stress was found ( $M=27.04$ ,  $SD= 6.13$ ,  $n=183$  for respondents who have children,  $M= 26.64$ ,  $SD= 5.43$ ,  $n=249$  for respondents who do not have children), ( $t(430)= 1.23$ ,  $p = 0.22$ ). Cohen's  $d$  also indicated a very small effect size ( $0.06$ ).

# Independent t-test in R

## Step 6

### Step 6: Report your findings

**Note: This does not reflect our example  
– it is for illustration only**

Statistically significant result (not our example):

An independent-samples t-test was conducted to compare perceived stress scores for respondents who have children and those who do not have children. Levene's test for equality of variances was not statistically significant, indicating that the assumption of homogeneity of variance was met ( $p > .05$ ). A statistically significant difference in the scores for perceived stress was found ( $M=31.20$ ,  $SD=5.79$ ,  $n=183$  for respondents who do not,  $M=33.64$ ,  $SD=5.30$ ,  $n=249$  for respondents who do have children), ( $t(431)=0.692$ ,  $p < 0.05$ ). A very small effect size was also indicated by the eta squared value (0.001).

# Independent t-test in R

## Step 6

### Step 6: Report your findings

**Note: This does not reflect our example  
– it is for illustration only**

Statistically significant result (with unequal variances):

A Welch's independent-samples t-test was conducted to compare perceived stress scores for respondents who have children and those who do not have children as Levene's test for equality of variances was statistically significant, indicating that the assumption of homogeneity of variance was not met ( $p < .05$ ). A statistically significant difference in the scores for perceived stress was found ( $M=31.20$ ,  $SD= 5.79$ ,  $n=183$  for respondents who do not,  $M= 33.64$ ,  $SD= 5.30$ ,  $n=249$  for respondents who do not have children), ( $t(431)= 0.692$ ,  $p < 0.05$ ). A very small effect size was also indicated by the eta squared value (0.001).

# Assumptions

---

The  $t$ -test is based on assumptions of

- Normality
  - The continuous variable of interest should follow a normal distribution
- Homogeneity of variance
  - The scatter for values of the variable for each of the groups should be similar
- You can test for both these
- BUT...
  - As long as the samples in each group are large and nearly equal, the  $t$ -test is robust
  - That is, still good, even though assumptions are not met.

# Probability and Statistical Inference

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DIFFERENCE (NON-PARAMETRIC)

MATH9102-W4-NONPARAM.QMD

A solid orange horizontal bar spanning the width of the slide at the bottom.

# Assumptions for Non- Parametric Tests

Random samples

Independent observations

- Each case is counted only once
- Unless it is a repeated measure where the same participants are retested on different occasions or under different conditions

# Non- Parametric Example from Andy Field

Our neurologist investigating the depressant effects of certain alcoholic drinks:

- Tested 20 students who went to a social event on Saturday night
  - 10 were allowed to drink only DrinkX
  - 10 were allowed to drink only DrinkY
- Levels of depression were measured using the Beck Depression Inventory (BDI) the day after and midweek.

**Field-BDI-NonParametric.csv**

# Non- Parametric Example from Andy Field

We are hypothesising that two groups of different entities will differ on the result of a test for depression.

A similar number of high and low ranks in each group would suggest depression levels do not differ between the groups.

How will we work it out?

- A greater number of high ranks in the DrinkX group than the DrinkY group would suggest the DrinkX group is more depressed than the DrinkY group.

# Non-Parametric: Comparing two independent conditions

This requires us to fit a model that compares the distribution of this variable for those in the DrinkY group and the DrinkX group.

- Requires us to test differences between two conditions in which different participants have been used.

To compare the distribution of two conditions and these conditions then you can use:

- The Mann-Whitney test and
- Wilcoxon's Rank-sum test

These work on ranking data.

## Ranking Data

The tests work on the principle of ranking the data for each group:

- Lowest score = a rank of 1,
- Next highest score = a rank of 2, and so on.
- Tied ranks are given the same rank: the average of the potential ranks.

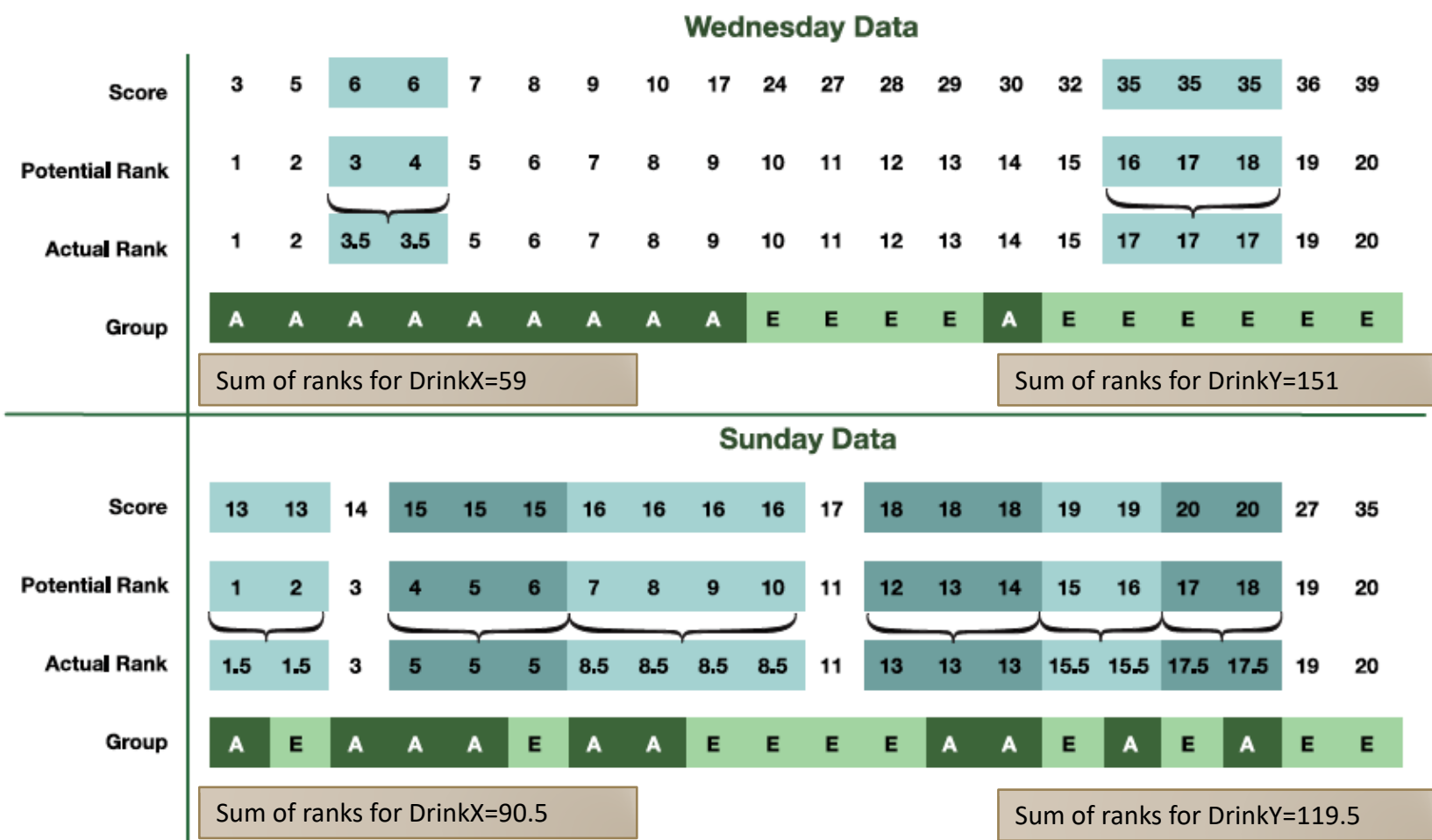
For an unequal group size the test statistic ( $W_s$ ) = sum of ranks in the group that contains the least people.

For an equal group size the test statistic  $W_s$  = the value of the smaller summed rank.

Add up the ranks for the two groups and take the lowest of these sums to be our test statistic.

The analysis is carried out on the ranks rather than the actual data.

# Ranking the Depression scores for Wednesday and Sunday



**FIGURE 6.4** Ranking the depression scores

# Provisional analysis using R

---

## Load the data

```
drinkset<- read.csv("Field-BDI-Non-parametric.csv",
header=TRUE)
```

## Get your descriptive statistics

- We know it is non-parametric so we want median and IQR

```
by(drinkset$bdisun, drinkset$drink, median)
```

```
by(drinkset$bdisun, drinkset$drink, IQR)
```

## Create your plots using ggplot

# Mann-Whitney test

---

It is used to test the null hypothesis that two samples come from the same population (i.e. have the same median)

- OR alternatively, whether observations in one sample tend to be larger than observations in the other.

Compares the medians from two populations and works when the  $Y$  variable is continuous, and the  $X$  variable is discrete with two attributes.

- Of course, the Mann-Whitney test can also be used for normally distributed data
  - But in that case, it is less powerful than the 2-sample  $t$ -test.

# The U Value

---

The U value for each group is calculated by subtracting the possible minimum rank which the group can take from the sum of the ranks, and the smallest U value is used for the test.

For the test of significance of the Mann-Whitney U-test it is assumed that with  $n > 80$  or each of the two samples at least  $> 30$  the distribution of the U-value from the sample approximates normal distribution.

U value is compared to table of critical values depending on the size of each sample

- U value must be *equal to or less than critical value*

# In R – Mann Whitney U (Wilcoxon Test)

---

From stats package

`wilcox.test(y~A)` #where `y` is your continuous variable and `A`

`is your categorical`

```
Test for differences on Sunday to get our U statistic (reported as W)
stats::wilcox.test(bdisun~drink, data=drinkset)
```

```
Warning in wilcox.test.default(x = c(15L, 35L, 16L, 18L, 19L, 17L, 27L, : cannot
compute exact p-value with ties
```

```
##
Wilcoxon rank sum test with continuity correction
##
data: bdisun by drink
W = 64.5, p-value = 0.2861
alternative hypothesis: true location shift is not equal to 0
```

# In R – Mann Whitney U (Wilcoxon Test)

---

From stats package

`wilcox.test(y~A)` #where `y` is your continuous variable and `A`

is your categorical

```
Test for differences on Wednesday to get our U statistic (reported as W)
stats::wilcox.test(bdiwed~drink, data=drinkset)
```

```
Warning in wilcox.test.default(x = c(28L, 35L, 35L, 24L, 39L, 32L, 27L, : cannot
compute exact p-value with ties
```

```
##
Wilcoxon rank sum test with continuity correction
##
data: bdiwed by drink
W = 96, p-value = 0.000569
alternative hypothesis: true location shift is not equal to 0
```

# Calculating an Effect Size

---

The equation to convert a z-score into the effect size estimate,  $r$ , is as follows (from Rosenthal, 1991: 19):

$$r = \frac{Z}{\sqrt{N}}$$

- $z$  is something we need to calculate if we use the stats package
- We can use `qnorm()` of the pvalue to get it or use the `wilcox_test` from the coin package

# In R – Calculate effect size

---

We can calculate our Z statistic by using the `wilcox_test` from the `coin` package

#To calculate Z we can use the Wilcox test from the `coin` package

```
coin::wilcox_test(bdiwed~as.factor(drink),
data=drinkset)
```

And calculate the effect size  $r$  by using the `wilcox_effsize` from the `rstatix` package

#Calculate the effect size  $R$

```
reff=rstatix::wilcox_effsize(bdiwed~drink,
data=drinkset)
```

# Reporting the Results

---

A Mann–Whitney test was used to test whether depression levels differed between those who drank DrinkX (Mdn = 17.50, IQR = 3.75) and those who drank DrinkY (Mdn = 16.00, IQR = 3.00) the day after the drinks were taken (Sunday). The results showed no significant difference ( $U = 64.5$ ,  $z = 1.11$ ,  $p = .29$ ,  $r = .25$ ).

However, a Mann–Whitney test used to compare depression levels three days after the drinks were taken (Wednesday) indicated that those who drank DrinkX (Mdn = 33.50, IQR = 6.75) were significantly more depressed than those who drank DrinkY (Mdn = 7.50, IQR = 3.75) ( $U = 96.00$ ,  $z = 3.48$ ,  $p < .001$ ,  $r = .78$ ).

## NOTE

- Our W statistic is what we use for U
- We report the median values because the data is not normally distributed.